MILITARY SPECIFICATION SHEET

ELECTRON TUBES, POWER

TYPES 5946, 6161

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, forced-air cooled
See figure 1
Mounting position: Any
Weight: 8 ounces (226 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	Fl MHz	Ef V (see note 2)	Eb Vdc	Ec Vdc	epy kv (see note 3)	egx v	Ib mAdc	Ic mAdc (see note 5)	ib a	ic a
TYPE 5946										
Anode pulsed osc and amp										
tp = 10 µsec (see note 1)										
Maximum:	1250	6.3±10%	---	---	7.5	600	45	10	4.5	1.0
Minimum:	---	---	---	---	---	---	---	---	---	---
tp = 100 µsec (see note 1)										
Maximum:	1250	6.3±10%	---	---	7.5	600	250	70	3.5	0.75
Minimum:	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3	---	---	---	---	---	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Pi W	Pp W	tk sec	T(grid seal, cathode) °C	T(rad) °C	Barometric pressure, reduced mmHg
TYPE 5946						
Anode pulsed osc and amp						
tp = 10 µsec (see note 1)						
Maximum:	340	250	---	150	180	141
Minimum:	---	---	60	---	---	---
tp = 100 µsec (see note 1)						
Maximum:	340	250	---	150	180	141
Minimum:	---	---	60	---	---	---
TEST CONDITIONS:	---	---	---	See note 4	See note 4	---

Ⓐ Denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

ABSOLUTE RATINGS:

Parameters: Unit:	Fl MHz	Ef V (See note 2)	Eb Vdc	Ec Vdc	epy kv (See note 3)	egx v	Ib mAdc	Ic mAdc (See note 5)	ib a	ic a
TYPE 6161										
C Telep: Anode modulated										
Maximum:	900	6.3±10%	1300	-300	---	---	210	75	---	---
Minimum:	---	---	---	---	---	---	---	--	---	---
C Telep and FM Telep										
Maximum:	900	6.3±10%	1600	-300	---	---	250	75	---	---
Minimum:	---	---	---	---	---	---	---	--	---	---
TEST CONDITIONS:	---	6.3	1600	---	---	---	---	--	---	---

ABSOLUTE RATINGS:

Parameters: Unit:	Pi W	Pp W	tk sec	T(grid seal, cathode) °C	T(rad) °C	Barometric pressure, reduced mmHg
TYPE 6161						
C Telep: Anode modulated						
Maximum:	270	167	---	150	180	141
Minimum:	---	---	60	---	---	---
C Telep and FM Telep						
Maximum:	400	250	---	150	180	141
Minimum:	---	---	60	---	---	---
TEST CONDITIONS:	---	---	---	See note 4	See note 4	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1201	Short and discontinuity detection		0.4	II	---	---	---	---
1301	Heater current	Preheat 1 minute	0.65	II	If	3.05	3.75	A
1261	Electrode voltage (grid)	Ec/Ib = 1.0 mAdc	0.65	II	Ec	---	-165	Vdc
1266	Total grid current	t = 120; Eb = 1,000 Vdc; Ec/Ib = 250 mAdc See note 6	0.65	II	Ic	---	-30	µAdc
1261	Electrode voltage (anode)	Ec = -10 Vdc; Eb/Ib = 250 mAdc	0.65	II	Eb	500	850	Vdc
Ⓐ 1231	Pulsing emission sinusoid	See note 7	0.65	II	is	12	---	a
	Quality conformance inspection, part 2							
1316	Amplification factor	Ec = -10 and -20Vdc; Eb/Ib = 250 mAdc; Mu = ΔEb/10	---	---	Mu	16	34	---
1236	Power oscillation (anode pulsed) (type 5946)	F = 1250 MHz; epy = 7.5 kv; Ib = 50 to 60 mAdc; Ic = 2 to 10 mAdc; Rk = 100 ohms; tp = 5 µsec; Du = 0.01 See note 8	---	---	Po	12	---	kW
1236	Power oscillation (type 6161)	Ic = +10 to +75 mAdc; Ek = 70 to 125 Vdc; F = 900 MHz; Eb = 1,670 to 1,725 Vdc; Ib = 250 mAdc See note 9	---	---	Po	90	---	W useful
1331	Direct-interelectrode capacitance	No shield No shield Shield See note 11	--- --- ---	--- --- ---	Cgp Cgk Cpk	5.6 10.5 0.12	6.6 12.5 0.26	pF pF pF

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 3							
Ⓐ ----	Life-test	Group D; power oscillation (anode pulsed) (type 5946) See notes 8 and 10	---	---	t	500	---	hr
		Group D; power oscillation (Type 6161) Ef (See notes 9 and 10)	---	---	t	500	---	hr
Ⓐ ----	Life-test end points:							
Ⓐ 1236	Power oscillation (Type 5946)	(anode pulsed)	---	---	Po	10	---	kW
Ⓐ 1236	Power oscillation (Type 6161)		---	---	Po	75	---	W (useful)
----	Vibration	No voltages applied See notes 12 and 13	---	---	---	---	---	---
1002	Barometric pressure, reduced	Eb = 1,600 Vdc; g and k tied together (voltage is between anode and grid); pressure = 141 mmHg See note 13	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

NOTES:

1. On time (tp) is defined as the sum of the duration of all the individual pulses during any 1,000 µs interval.

2. Because the cathode is subjected to considerable back bombardment as the frequency is increased with resultant increase in temperature, the heater voltage shall be reduced depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

3. The magnitude of any spike on the anode voltage pulse should not exceed a value of 8,500 volts with respect to cathode and its duration should not exceed 0.5 microseconds measured at the peak-pulse-value level.

4. The air flow must be adequate to limit the temperature of the radiator, terminals, and seals to their respective maximum values. The radiator temperature should be measured at point 1, figure 4; the seal temperature should be measured at point 2, figure 4. Air flow should begin before the application of any voltage. Heater power, anode power, and air may be removed simultaneously.

5. Range of rated grid current versus operating frequency chart (see figure 5). Maximum negative grid current shall not exceed 10 milliamperes at all frequencies.

6. Airflow shall be supplied to limit the radiator temperature to 160° to 180°C (point 1, figure 4).

7. There shall be no signs of arcing or gas.

8. The 5946 tube shall be tested in a pulsed oscillator circuit (see figure 6). Peak power output is obtained by dividing the average power output by the output duty factor (D4). If sparking occurs, the tube shall be operated for 5 minutes. During this period there shall be no more than 10 sparks/min.

A 9. The 6161 tube shall be tested in a grounded grid oscillator circuit (see figure 7) with anode to grid voltage 1,670 to 1,725 Vdc and cathode to grid voltage of 70 to 125 Vdc. An Rg1 grid resistor of 2,000 ohms (max) may be used to stabilize operation.

10. Life test

    Heater voltage for Type 6161 shall be initially set at 6.3 volts and then reduced to 5.5 volts after other test conditions have been achieved. For Type 5946 set ef = 6.3 V. Seal temperatures shall not exceed 150°C during life, as measured at point 2, figure 4.

11. Use flat metal shield 7-1/2 inch outside diameter connected to grid terminal.

12. Vibration shall be at 50 Hz and 10G for a period of 2 minutes in direction perpendicular to the axis of the tube and for 2 minutes in a direction parallel to the axis of the tube. The tube shall be clamped firmly by the anode ring. The tube shall pass all electrical tests after vibration.

13. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AOL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

A 14. Changes from Previous Issue The margins of this revision are marked with a letter A to indicate where changes from the previous revision have been made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this revision without regard to the marginal notations and the relationship to the previous issue of this document.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Preparing activity:

Air Force - 85

Review activities:

Army - EL
Navy - SH
Air Force - 99
DSA - ES

(Project No. 5960-3023)

User activities:

Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

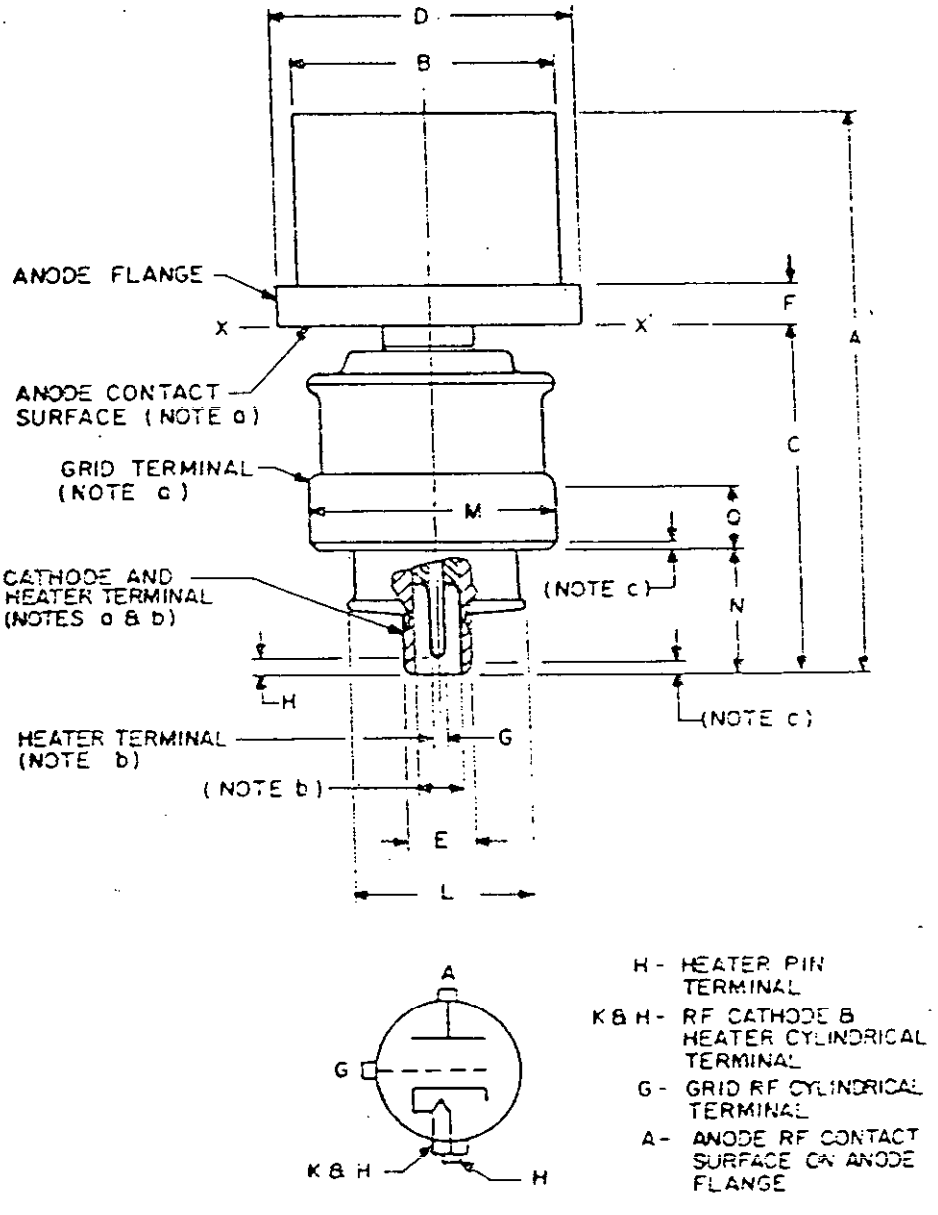


FIGURE 1 Outline drawing of electron tube types 6161 and 5946

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

LTR	Dimensions in inches - metric equivalents (mm; in parentheses)		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 1			
H		.062 (1.57)	
Quality conformance inspection, part 2			
A	3.219(81.76)	3.406(86.51)	
B	1.484(37.69)	1.516(38.51)	
C	2.000(50.80)	2.125(53.98)	
D	1.740(44.20)	1.750(44.70)	d
E	370(9.40)		
M	1.358(34.75)		
N	719(18.26)	.781(19.84)	
Quality conformance inspection, part 3			e
F	.219(5.56)	.281(7.14)	
G	.053(1.35)	.061(1.55)	
L		1.125(28.58)	
Q	.312(7.92)		

NOTES:

a) With the cylindrical surfaces of its grid and cathode terminals clean, smooth, and free of burrs, the tube shall enter a gage as shown in Figure 2. The four cylindrical holes H1, H2, H3, and H4 have axes coincident within .0005 (.01). Lengths shall be determined from Figure 1), and successively smaller diameters as shown in Figure 2. The anode flange shall be entirely engaged by hole H1 and the contact surface of the anode flange shall seat on the shoulder between holes H1 and H2. The plane surface of this shoulder is 90° ± 2 minutes to the axes of the holes. Seating is determined by failure of a .005 (.13) thickness gage, .125 (3.12) wide to enter more than .062 (1.57) between the shoulder surface and the anode contact surface. With the tube properly seated as described herein, the grid terminal shall be entirely engaged by the hole H3, and the cathode terminal shall be engaged by hole H4 to a depth of at least .250 (6.35).

b) Concentricity of the heater terminal with respect to the cathode terminal shall be determined by a gage as shown in Figure 3. The cylindrical hole H5 and the annular holes, H6-H7 have axes coincident within .0005 (.01). The cathode terminal and the heater terminal shall enter this gage to a depth of .375 (9.53).

c) May be round or beveled, not to exceed .062 (1.57).

d) The average of the minimum reading and the reading 90° from it shall be equal to or greater than 1.740 (44.20). The difference between the two readings shall not exceed .010 (.25) for any one tube.

e) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1 Outline drawing of electron tube types 5946 and 6161.
(Continued)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

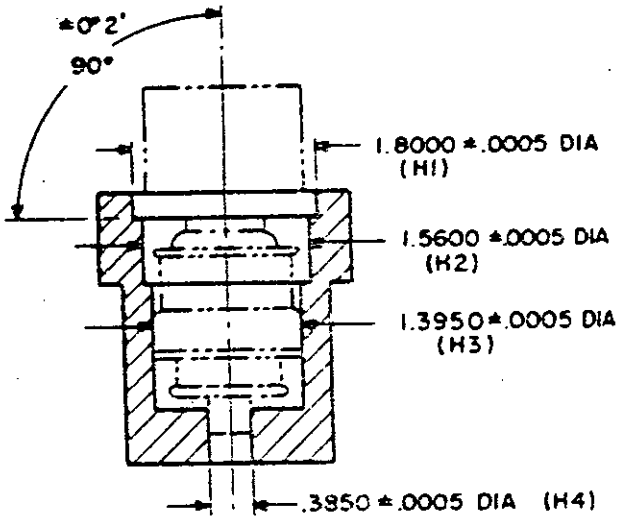


FIGURE 2 Anode, grid, cathode gage

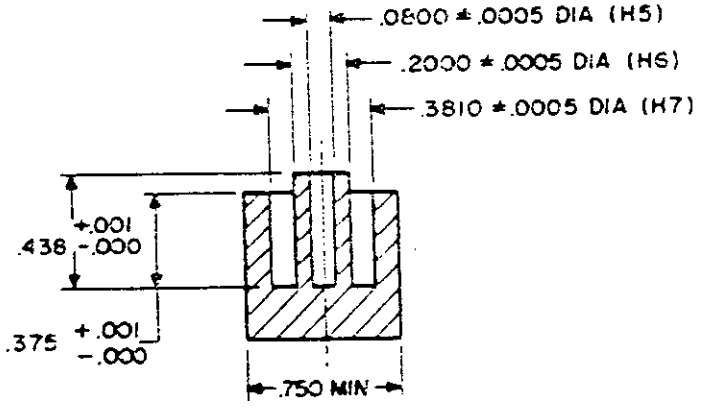


FIGURE 3 Heater and cathode terminal gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

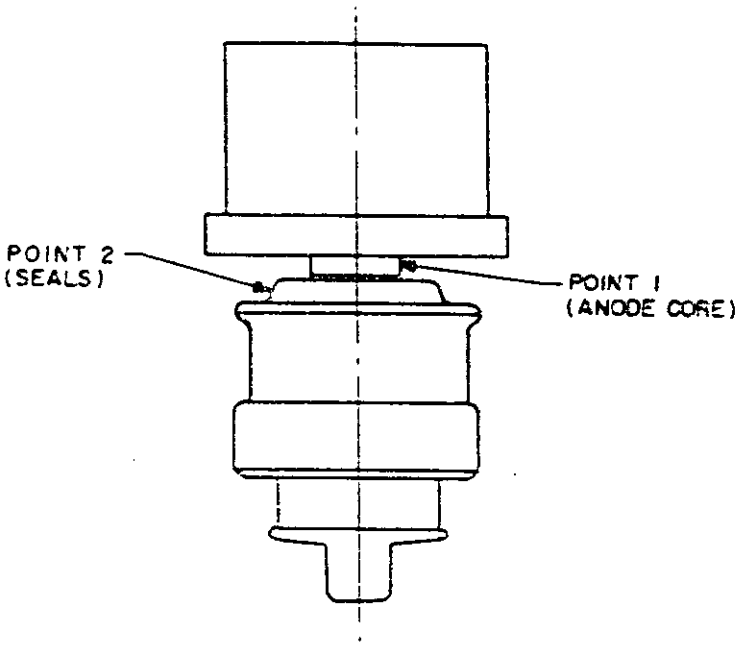


FIGURE 4 Temperature measuring locations

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

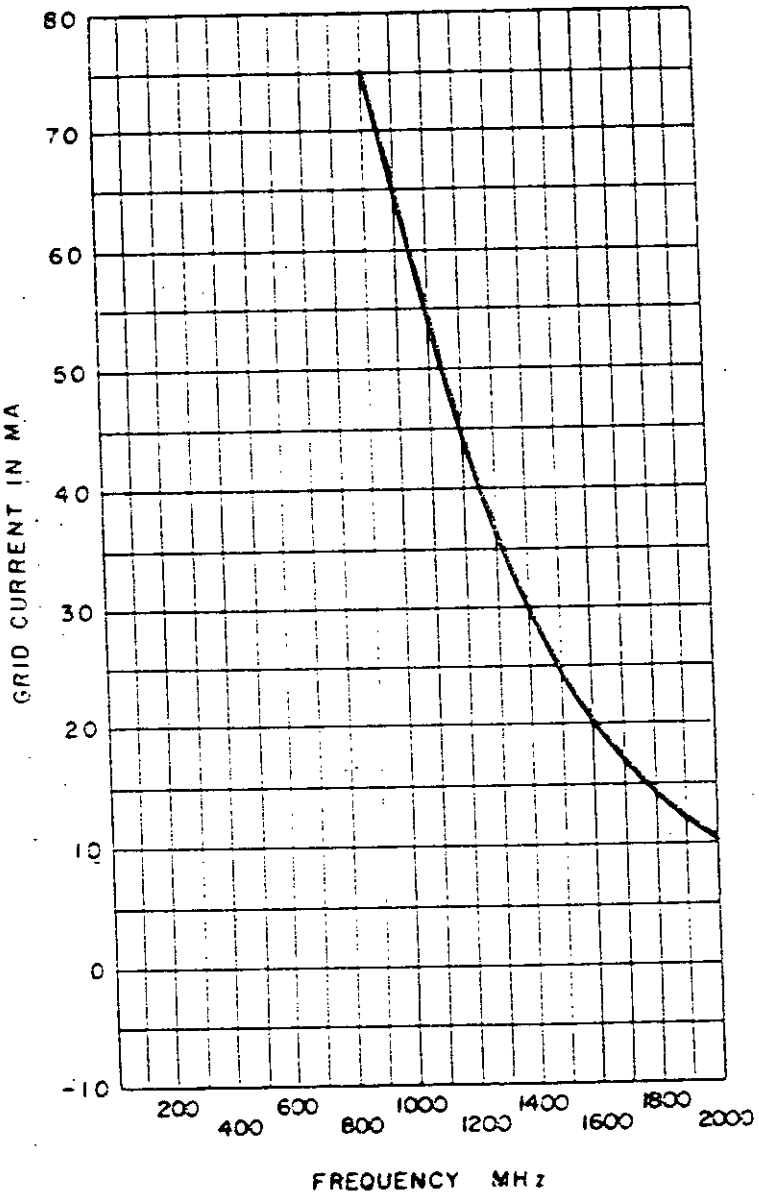
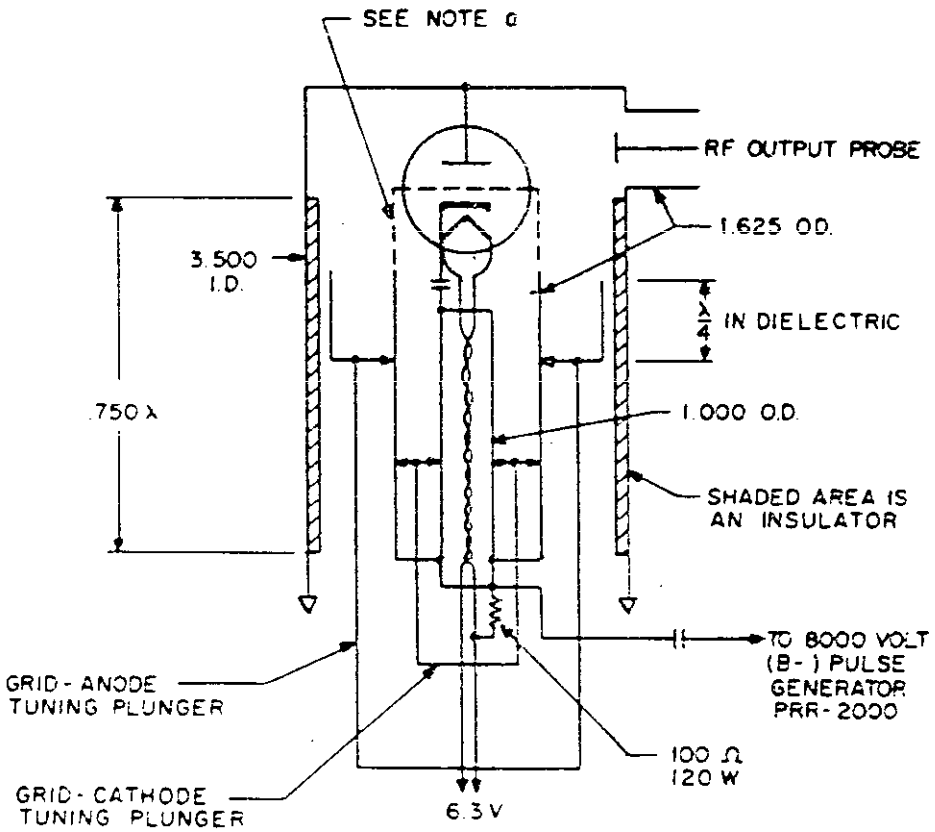


FIGURE 5 Chart, grid current versus operating frequency

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.



NOTE
a) Grid line slotted for feedback

FIGURE 6 1250 MHz pulsed oscillator circuit for type 5946

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

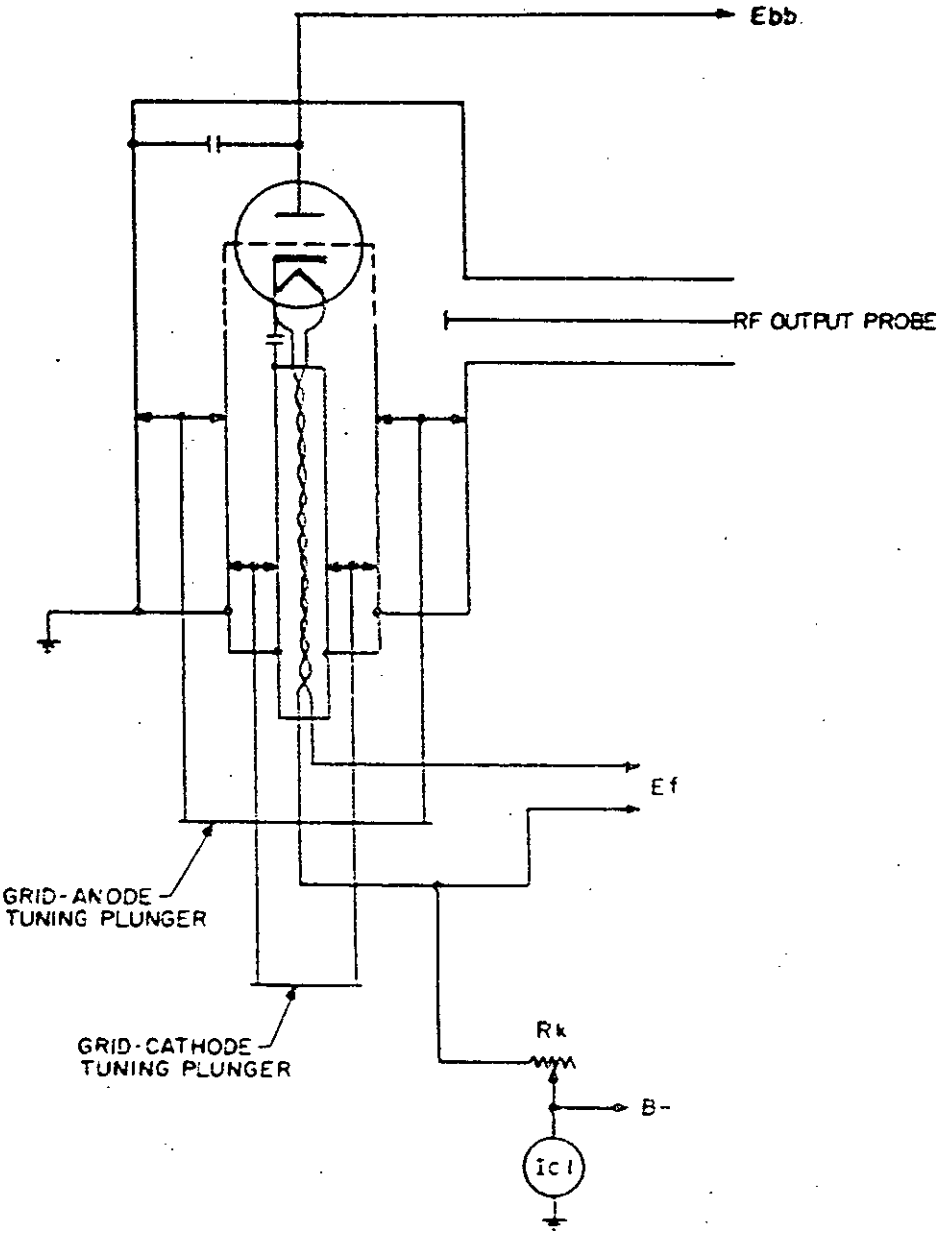


FIGURE 7 900 MHz oscillator circuit for type 6161

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.